

# The Arrow's Engine: The Dynamical Rule of Event-Density Gravity, and the Closing of Its Weak-Field Assumptions

Allen Proxmire

June 2026

## Abstract

This paper closes the two weak-field assumptions of GR-I, builds the dynamical rule GR-II reduced the field equation to, and resolves KM-II §6's open scalar-mode question. GR-I derived Event Density's weak-field Einstein metric under two labeled assumptions — the commitment-rate linearity  $\alpha = 1$  (selecting the Einstein over the Nordström branch) and the timelike geodesic identity (that massive matter free-falls on the bandwidth metric). GR-II established that ED's gravity is khronometric and reduced the field equation, by Lovelock's theorem, to *the steady state of an admissible dynamical-bandwidth rule*  $\dot{b} = F$ , leaving  $F$  unbuilt and the khronon's scalar-mode speed open (KM-II §6). This paper builds  $F$ , closes both assumptions, and fixes the khronon speed — with one spine running through all of it: **the arrow (P11 irreversibility) selects the Einstein/khronometric structure three independent times**. (i)  $\alpha = 1$  is **forced** from the P04 band law: the commitment-reserve is a distinct, *monotone-draining* band, and the Nordström branch would require it to replenish in step with the metric band — which P11 forbids; Nordström is positively excluded (modulo a named band-accounting premise). (ii) **The timelike geodesic identity is reduced** to the massive eikonal of GR-I's proven null result — the massive Jacobi functional is limit-forced to recover both the null (Fermat) and Newtonian (Maupertuis) limits, with the explicit  $\Sigma$ -rule computation the residual. (iii) **The khronon propagates at the speed of light** ( $c_s = c$ ): the only candidate for a foliation-specific kinetic term that would give it a second cone is the reserve sector, and P11 makes that sector dissipative — it *damps* the khronon (confirmed in simulation: it damps and overdamps without shifting the cone), it does not supply the conservative  $\lambda\theta^2$  term a second cone needs; this resolves KM-II §6 toward maximal predictivity. The rule is then **built and run**: in its forced form ( $\dot{b} = D\nabla^2 b - \kappa\rho$ , P02 adjacency-sharing minus P11 matter-concentration sink) its steady state is the Newtonian field equation  $\nabla^2 b \sim \rho$  (the fixed point of the two forced terms); the deficit amplitude is exactly linear in source (Schwarzschild  $r_s \propto M$ , confirmed in 2D and 3D); strong coupling forms a **finite-radius**  $b \rightarrow 0$  **horizon** that is a **frozen A2 decoupling cut** — realizing dynamically the identity that the A2 cut, the metric horizon, and the  $b \rightarrow 0$  locus are one object; the emergent horizon carries **area-law (holographic) entropy**  $S \propto A$ ; and the hyperbolic rule carries **2 tensor + 1 scalar** modes (the mode count read off  $F$  rather than the gauge group). The emergent horizon's thermodynamics follow: area-law  $S \propto A$  (measured) and the Hawking scaling  $\kappa = 1/(2r_s) \propto 1/r_h$  (derived from the vacuum solution  $b = 1 - r_s/r +$  GR-I's  $g_{00}g_{rr} = -1$ , with  $r_s \propto M$  measured). One number remains honestly open and is flagged in-paper: the preferred-frame parameters  $\alpha_1, \alpha_2$  (ED is driven into the maximally-suppressed

corner by five independent mechanisms, but the *values* are not computed and not faked). The result: **the rule ED’s gravity reduces to is built, its weak-field assumptions are closed, and its scalar mode is pinned to the light cone — with the same irreversibility doing all three jobs.** Einstein is not derived; one falsification number  $(\alpha_1, \alpha_2)$  remains open; the engine exists and runs.

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## 1. Introduction

### 1.1 What this paper does

GR-I and GR-II took ED’s gravity to the khronometric class with the weak-field Einstein tests — but on two labeled assumptions and with the central object, the dynamical rule, unbuilt. This paper closes that. It (a) **forces**  $\alpha = 1$  from the P04 band law (§4), (b) **reduces the timelike geodesic identity** to a proven limit (§5), (c) **derives the khronon speed**  $c_s = c$  (§6, resolving KM-II §6), and (d) **builds and runs the dynamical rule**  $F$  (§7), confirming the Newtonian field equation, the Schwarzschild mass-scaling, an emergent frozen horizon, the area law, the Hawking scaling, and the mode count. It then states honestly the **one number that remains open** (§8): the preferred-frame  $\alpha_1, \alpha_2$  (the Hawking scaling, earlier thought open, is corrected there to a derived result).

### 1.2 Why this matters, and the register

The register of this paper is *closing the gaps*, not breakthrough. Its value is that the quartet’s two standing weak-field assumptions become derived (or reduced), the object the whole construction reduces to is shown to exist and run, and a published open question (KM-II §6) is answered. The conceptual spine is worth stating plainly: **one primitive — the arrow, P11 irreversibility — selects the Einstein/khronometric structure three separate times** (the lapse, the mode count, the scalar speed). That the factor of two, the extra scalar, and that scalar’s speed all trace to one commitment is the paper’s organizing fact.

### 1.3 How this fits the arc

- **GR-I:** the weak-field metric and classical tests, on the  $\alpha = 1$  and geodesic assumptions. (*this paper closes both*)
- **GR-II:** the khronometric class; the field equation reduced to the steady state of  $F$  (unbuilt); the scalar speed open. (*this paper builds  $F$ ; fixes the speed*)
- **KM-I/KM-II:** the deep-field (MOND) and cosmological sectors, inheriting the weak-field metric. (*KM-II §6 resolved here*)
- **GR-III (this paper):** the engine — the dynamical rule built, the weak-field assumptions closed, the khronon speed pinned, one number  $(\alpha_1, \alpha_2)$  left open.

### 1.4 Conventions and regime of validity

Signature  $(-, +, +, +)$ ;  $c = 1$  where convenient. The derivations are **weak-field / DCGT-limit**; the simulations are minimal-model (modest grids; coefficients  $\kappa, D$  unpinned). Where a result is *measured* it is labeled so; where *forced/derived/reduced* it is tiered; where *open* it is flagged. No strong-field claim is made beyond the emergent-horizon structure; the  $\alpha_1, \alpha_2$  values

are explicitly deferred, and the Hawking scaling is derived (the rule’s vacuum solution + GR-I) with the direct sim measurement resolution-limited.

## 2. Primitive Inputs

**Substrate primitives (postulated, Paper\_087), load-bearing:** P11 (commitment irreversibility — the arrow; load-bearing three times), P13 (the tick), P04 (the four-band bandwidth, including the commitment-reserve band), P02 (reciprocal adjacency — the shared metric band), P05 (transport — the single causal cone).

**Inherited results:** GR-I ( $g \sim b^{-1}$ ,  $N^2 \sim b$ , the weak-field metric; the  $\alpha = 1$  and geodesic assumptions this paper closes); GR-II (the khronometric class; the field equation = steady state of  $F$ ; the mode-count gauge argument); Paper\_027 ( $G$ , value-inherited); R2 (the dynamical rule’s admissibility); R4–R9 (the structural arc: genuine metric, conserved source, Lovelock).

**External dictionary (inherited, labeled I):** Lovelock’s theorem (GR-II); the khronometric/Einstein-aether mode and PPN structure; the 3D-vs-2D CFL stability bound for explicit diffusion.

**No new primitive is introduced; no primitive forcing is invoked.** The dynamical rule is the formalization of the commitment-reserve dynamics P04 + P11 already declare (R2).

### 2.5 Load-Bearing Step Audit

*Status legend:* **I** = inherited; **D** = derived here; **D-cond** = derived modulo a named premise/limit; **measured** = simulation result; **structural** = from structure alone; **fixed-point** = the steady state of the forced terms (built-in via the terms being primitive-grounded); **open** = declared open, not computed.

| Step                                                                         | Status                                                       | Source   |
|------------------------------------------------------------------------------|--------------------------------------------------------------|----------|
| field equation = steady state of $\dot{b} = F$                               | <b>I</b>                                                     | GR-II R9 |
| $\alpha = 1$ forced (Nordström excluded)                                     | <b>D-cond</b> (band-accounting premise)                      | §4       |
| Nordström needs reserve replenishment $\rightarrow$ inadmissible (P11)       | <b>structural</b>                                            | §4       |
| timelike geodesic identity = massive eikonal of the null result              | <b>D-cond</b> (limit-forced; $\Sigma$ -computation residual) | §5       |
| $c_s = c$ ( $\varepsilon = 0$ , khronon at light speed)                      | <b>D</b> (leading order)                                     | §6       |
| reserve dissipative $\rightarrow$ damps, not a $\lambda\theta^2$ second cone | <b>structural + measured</b>                                 | §6       |
| the arrow (P11) selects the structure three times                            | <b>structural unification</b>                                | §4–§6    |

| Step                                                           | Status                                                                                           | Source     |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------|
| $F$ built; steady state = $\nabla^2 b \sim \rho$               | <b>fixed-point</b> (corr 0.999)                                                                  | §7         |
| $r_s \propto M$ (Schwarzschild mass-scaling), 2D + 3D          | <b>measured</b>                                                                                  | §7         |
| finite-radius $b \rightarrow 0$ horizon = frozen A2 cut        | <b>measured (emergent)</b>                                                                       | §7         |
| $S \propto A$ (area-law / holographic) on the emergent horizon | <b>measured</b>                                                                                  | §7         |
| 2 tensor + 1 scalar, from $F$                                  | <b>D</b>                                                                                         | §7         |
| $\alpha_1, \alpha_2$ values                                    | <b>open</b> — structural suppression only, not computed                                          | §8         |
| Hawking $T \propto 1/r_h$ ( $\kappa = 1/(2r_s)$ )              | <b>derived</b> (vacuum solution + GR-I; $r_s \propto M$ measured); direct sim resolution-limited | §8         |
| full EFE / strong field / pure GR                              | <b>NOT CLAIMED</b>                                                                               | preamble 2 |

### 3. The Rule the Field Equation Reduces To

GR-II’s Lovelock argument (R9) established that ED’s field equation is **the steady state of an admissible dynamical-bandwidth rule**  $\dot{b} = F(\rho, \nabla\rho, b)$  — the per-edge bandwidth evolution. R2 showed this rule is admissible: it is the formalization of the commitment-reserve dynamics P04 + P11 already declare, violating no certified invariant, in its commitment-driven, monotone-conservative, orientation-blind form. The structural arc then *forced its form*: the band↔edge map (R5, by flux↔Levi-Civita covariance), the lapse exponent ( $\alpha = 1$ , §4 below), and the geodesic structure (§5). What remained was to **build it, run it, and read off its modes** — and to fix the one open coefficient of its propagating sector, the khronon speed.

This paper does those four things. The rule’s minimal forced form is

$$\dot{b} = D\nabla^2 b - \kappa\rho, \quad b \geq 0, \quad b \rightarrow 1 \text{ at the frame,}$$

with  $D\nabla^2 b$  the **P02 adjacency-sharing** (the reciprocal metric band equilibrates by the graph Laplacian — the elliptic geometry sector) and  $-\kappa\rho$  the **P11 commitment-concentration sink** (persistent matter holds bandwidth in its concentrated channel, depleting the metric band  $\propto$  its density). The hyperbolic (propagating) sector replaces the relaxational  $D\nabla^2 b$  with the retarded single-P05-transport wave operator (§6–§7).

### 4. Closing the Lapse Assumption: $\alpha = 1$ Forced

GR-I’s lapse  $N^2 \sim b$  (hence the Schwarzschild relation  $g_{00}g_{rr} = -1$ , the Einstein branch) rested on  $\alpha = 1$  in  $\Gamma_{\text{commit}} \sim b_{\text{int}}/\text{reserve}$ . With  $\Gamma \sim b^\alpha$ , the null condition gives  $g_{00}g_{rr} \sim$

$-b^{2(\alpha-1)}$ :  $\alpha = 1$  is Einstein,  $\alpha = 0$  is Nordström (conformal, no light bending, experimentally excluded). Writing  $reserve \sim b_{int}^\beta$ ,  $\alpha = 1 - \beta$ : **the Einstein/Nordström fork reduces to a single question — whether the commitment-reserve band co-scales with the internal (metric) band.**

It does not, and P11 is why. The reserve and  $b_{int}$  are **different state variables**:  $b_{int}$  is the *ambient* adjacency capacity (shaped by  $\rho$ ,  $\nabla^2 b \sim \rho$ ); the reserve is a *carried, monotone-draining budget* set by commitment **history** (P11; no replenishment, R2 §5). Generically, then,  $\beta = 0$ ,  $\alpha = 1$ . The Nordström value  $\beta = 1$  requires the reserve to *track*  $b_{int}$  — to be **raised** where  $b_{int}$  is high — which is **replenishment**, forbidden by the same P11 irreversibility that defines commitment.

**The P04 band law forces  $\alpha = 1$  and positively excludes Nordström.** The Einstein branch at the lapse is selected by the arrow (P11): the reserve cannot replenish to track the metric band, so it does not co-scale, so  $\alpha = 1$ . (*Modulo one band-accounting premise: that the metric band and the rate-numerator band are the same  $b_{int}$ ; named, not proven.*) Had P04 defined the reserve as a fixed fraction of the total or supplied a replenishment band, ED would have been Nordström — experimentally excluded. It does neither. **This is the arrow’s first hat.**

GR-I’s labeled postulate P-Commitment-Linear is thereby a derived result.

## 5. Closing the Geodesic Assumption: the Timelike Identity Reduced

$\nabla \cdot T = 0$  for ED’s matter reduces (R6) to  $\rho a^\nu$  — so conservation holds iff the bandwidth worldlines are geodesics of  $g \sim b^{-1}$ . GR-I proved this for **null** fronts (Fermat, optical index  $n_{opt} \sim b^{-1}$  — the source of the factor of two); the **timelike/massive** case was assumed.

The timelike-geodesic functional for  $g \sim b^{-1}$  is the **Jacobi/Maupertuis** function  $J(b) = b^{-1} \sqrt{E^2 - m^2 b}$ , and it is *limit-forced* — it reduces to a *proven* result in each of the two checkable limits:

| limit             | J(b) reduces to          | interpretation                                                                         |
|-------------------|--------------------------|----------------------------------------------------------------------------------------|
| $m \rightarrow 0$ | $E/b = E \cdot n_{opt}$  | <b>Fermat</b> (the proven null result — so the timelike functional <i>contains</i> it) |
| weak field        | $\sqrt{2m(E_{kin} - V)}$ | <b>Maupertuis</b> (Newtonian orbits — recovers R1)                                     |

ED’s subluminal max- $\Sigma$  front is then identified as the **massive eikonal** of the same propagation GR-I treated masslessly (its Hamilton–Jacobi ray being the timelike geodesic; the worldline’s intrinsic clock from P11 + P13).

**The timelike geodesic identity is advanced from assumed to limit-forced reduction** — forced to recover both the proven null limit and the Newtonian limit, the two checks any correct functional must pass (and it could have failed the  $m \rightarrow 0$

check; it did not). The residual is the explicit  $\Sigma$ -rule massive-front computation; the explicitly-assumed identity is replaced by a located, limit-checked reduction.

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## 6. The Khronon Speed: $c_s = c$ , Derived (KM-II §6 Resolved)

GR-II's mode count (2 tensor + 1 scalar = khronometric) left the **scalar (khronon) speed** open (KM-II §6: “may propagate at a different speed in principle”). The hyperbolic rule reduces this to one coefficient: writing the propagating perturbation as  $\ddot{h}_{ij} = c^2 \nabla^2 h_{ij} + \varepsilon \cdot (\text{trace})$ , the khronon speed is  $c_s/c = \sqrt{1 + \varepsilon}$ , with  $\varepsilon$  a foliation-specific  $\lambda\theta^2$  kinetic term.  $\varepsilon = 0 \rightarrow$  khronon at light speed;  $\varepsilon \neq 0 \rightarrow$  a second cone.

$\varepsilon$  is derived to be zero. The only candidate source of a  $\lambda\theta^2$  term is the commitment-reserve sector — and P11 makes that sector **one-way / dissipative** (monotone drain, no replenishment). A dissipative coupling enters the dispersion as the **imaginary** part of  $\omega$  (damping), never the **real** part (speed); a conservative  $\lambda\theta^2$  term shifts the real part. So P11 irreversibility makes the reserve **incapable** of being the kinetic term  $\varepsilon \neq 0$  requires. Simulation confirms the distinction is physical: a dissipative reserve term **damps and overdamps** the khronon without moving its cone (the speed is unshifted, the amplitude decays), in sharp contrast to the  $\varepsilon$ -knob, which cleanly shifts  $c_s/c = \sqrt{1 + \varepsilon}$ . The forced rule (P05 transport + P11 drain) supplies no  $\lambda\theta^2$  term, and minimality forbids positing one.

$\varepsilon = 0$ ,  $c_s = c$  — **derived: ED's scalar gravitational-wave polarization is at the speed of light**, damped near active matter (overdamped where commitment fires) and clean in vacuum (observationally consistent — the far-field khronon is at  $c$ , undamped). This resolves KM-II §6 *as a derivation*, toward maximal predictivity — sharper than generic khronometric gravity, which allows  $c_s \neq c$ . **This is the arrow's third hat** (the second being the mode count: the arrow breaks the time-gauge that would freeze the scalar, GR-II/R10).

A scalar gravitational-wave polarization *at  $c$*  is, in principle, **observationally distinguishable** from both pure General Relativity (two tensor modes, no scalar) and generic khronometric gravity (a scalar at  $c_s \neq c$ ) — a sharp, specific signature of ED's foliation sector.

**The three hats.** One primitive, P11, selects the Einstein/khronometric structure three independent times:  $\alpha = 1$  at the lapse (the reserve cannot replenish — §4), the **khronon physical** at the mode count (the arrow breaks the time-gauge — GR-II), and the **khronon at light speed** (the reserve is dissipative, not kinetic — this section). The factor of two, the extra scalar, and that scalar's speed are one irreversibility wearing three hats. *The same arrow that makes the khronon keeps it on the light cone.*

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## 7. The Engine, Built and Run

The rule is built and run (`evaluation/DynamicalBandwidth/`); results separated into *built-in* (the fixed point of the forced terms) and *emergent* (not built in).

### 7.1 The Newtonian field equation (fixed point)

The steady state satisfies  $\nabla^2 b \sim \rho$  ( $\text{corr}(\nabla^2 b, \rho) = 0.999$ , the bandwidth-Laplacian concentrated  $52\times$  on the source, harmonic vacuum). This confirms the two forced terms (P02 sharing + P11 sink) are Newtonian-consistent — the content being that they are primitive-grounded, not that Poisson is a surprise.

### 7.2 Schwarzschild mass-scaling $r_s \propto M$ (emergent)

The deficit amplitude is **exactly linear** in the integrated source, in **both 2D and 3D** (3D:  $r_s = 0.158, 0.316, 0.633$  for  $M = 504, 1008, 2016$ ).  $r_s \propto M$  — the Schwarzschild relation. The raw 3D  $1/r$  slope ( $-1.68$ ) is finite-box/convergence-limited: **the mass-scaling is the invariant; the raw slope is not.**

### 7.3 The emergent horizon = a frozen A2 cut (the emergent result)

At strong coupling,  $b \rightarrow 0$  on a **finite-radius surface** ( $g_{rr} \sim 1/b \rightarrow \infty$ ) — a metric horizon — where the reserve is exhausted, so the cut is **frozen** (P11). Nothing in the rule mentions a horizon; one forms. This realizes dynamically the identity that the  $b \rightarrow 0$  locus is *at once* an **A2 decoupling cut**, a **metric horizon**, and a **V5 surface** — one rule, one locus, three identities. **A2-dynamical is hereby realized by build-and-run, not argued.**

### 7.4 Area-law entropy $S \propto A$ (emergent, holographic)

The severed-information count across the frozen cut (A1: capacity-zero across the cut  $\rightarrow$  the hidden DOF are the boundary adjacency channels) scales as the horizon **perimeter** ( $r_h^{0.96}$ ), **not** the enclosed bulk ( $r_h^{2.01}$ ). The entropy is a **surface** quantity — holographic — on a horizon the rule formed dynamically. (*Half-structural: A1 severance is a surface cut. Half-measured: the emergent horizon is a clean compact surface — the part that could have failed and did not. The  $1/4$  coefficient is value-inherited.*)

### 7.5 The mode count, from $F$ (derived)

The hyperbolic rule's gauge structure gives **2 tensor + 1 scalar** — the scalar physical because the arrow breaks the time-gauge — reproducing GR-II's khronometric count by direct operator analysis rather than the gauge-group argument. Tensor modes at  $c$  (single P05 cone); scalar at  $c$  (§6).

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## 8. The Open Edge — One Number Open, One Corrected

Two quantities were carried as open in earlier drafts of this work; re-examination resolves one (the Hawking scaling — a measurement correction) and leaves one genuinely open ( $\alpha_1, \alpha_2$ ). The paper states the open one as open rather than leaning on the structural pressure.

$\alpha_1, \alpha_2$  (**the preferred-frame front**) — **structurally suppressed, not computed**. ED is driven into the maximally-suppressed corner of the preferred-frame parameter space by **five independent derived mechanisms**: universal (not differential) Lorentz violation makes  $\alpha_1, \alpha_2$

sub-leading (GR-II);  $c_T = c$  and  $c_s = c$  put both gravitational cones at light speed (the most-suppressed khronometric regime); the metric-only matter coupling removes the direct khronon-matter  $\alpha_1$  source (KM-I); and — ED-specific — the **dissipative near-field** (the overdamped khronon around matter, §6) cannot build the frame-dependent near-field response  $\alpha_1, \alpha_2$  measure (P11, again). Five independent mechanisms point toward the same suppressed corner: strong pressure toward PPN-safety. **But the *values* are not computed here, and not faked** — they require the pinned couplings  $(\kappa, D)$  plus the khronometric PPN formulas, or a direct PPN expansion of  $F$ . The falsification front stays **formally open as a number**; what has changed is that the computation is now definite and foliation-knob-free.

**The Hawking temperature scaling  $T \propto 1/r_h$  — follows (derived; direct sim resolution-limited).** The emergent horizon carries both the area law ( $S \propto A$ , §7) and the Hawking scaling. The surface gravity, given GR-I’s Schwarzschild relation  $g_{00}g_{rr} = -1$  (metric  $-b dt^2 + b^{-1} dr^2$ ), is  $\kappa = \frac{1}{2}\partial_r b$  at the horizon; on the rule’s *vacuum* solution  $b = 1 - r_s/r$  (the linear rule is 3D-harmonic in vacuum) the horizon sits at  $r = r_s$  and  $\kappa = 1/(2r_s)$ , so with  $r_s \propto M$  measured,  $\kappa \propto 1/r_h$  — the Hawking relation. (*An earlier round reported this “flat” — a measurement error: it used  $\partial_r \sqrt{b}$ , which **diverges** at the horizon since  $\sqrt{b} = N \rightarrow 0$ ; the corrected  $\kappa = \frac{1}{2}\partial_r b$  is finite and scales. The minimal rule carries the Hawking scaling; no hyperbolic strong-field rule is required.*) The honest residual is that a **fully-direct** simulation of the near-horizon slope is resolution-limited — the elliptic relaxation smooths the  $b = 0$  clip over a  $D$ -set width, capping the *measured* slope at the small horizons compact sources allow — so  $\kappa \propto 1/r_h$  is tiered as **derived** (rule vacuum solution + GR-I relation + measured  $r_s \propto M$ ), not as a clean direct measurement. The thermodynamic coefficient ( $T = \kappa/2\pi$ ,  $S = A/4$ ) stays value-inherited.

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## 9. Position Relative to the Quartet

- **GR-I’s two assumptions are closed.**  $\alpha = 1$  is forced (§4); the timelike geodesic identity is limit-forced (§5). GR-I’s preamble items 6 and 9 are downgraded from postulate/assumption to derived/reduced, citing this paper.
- **GR-II’s reduced object is built.** The field equation = steady state of  $F$  (R9) is now an  $F$  that exists and runs (§7); the mode count is reconfirmed from  $F$  (§7); the field-equation form’s Newtonian limit is measured.
- **KM-II §6 is resolved.** The scalar-mode speed, left open, is  $c_s = c$  (§6, derived). KM-II’s §6 caveat becomes “the khronon is at light speed, derived.”
- **A2-dynamical (the working-program target #8) is realized.** The same rule that builds the EFE horizon forms A2’s frozen cut — by build-and-run (§7).

What this paper does **not** change: the gravitational class (khronometric, GR-II); the MOND/cosmology sectors (KM-I/II); the value-inherited constants; and the one open number,  $\alpha_1, \alpha_2$  (§8).

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## 10. The Wedge — Empirical Content

1. **The khronon at light speed** — a falsifiable, sharp prediction: ED’s scalar GW polarization propagates at  $c$  (not a fast/slow second cone), damped near matter, clean in vacuum. A scalar polarization at  $c$  is distinguishable in principle from pure-GR (two tensor modes only) and from generic khronometric gravity ( $c_s \neq c$ ).
  2. **The emergent horizon’s area law** —  $S \propto A$  on a dynamically-formed horizon; the holographic scaling is a structural prediction, the  $1/4$  value-inherited.
  3. **The preferred-frame numbers** —  $\alpha_1, \alpha_2$ , once computed (the definite, knob-free calculation §8 sets up), are the GR-II falsification front; this paper sharpens the lean and defines the computation, it does not pre-empt the verdict.
  4. **Not claimed as wedges:** the  $\alpha_1, \alpha_2$  values; the Hawking  $T$  coefficient (the scaling is derived; the  $T = \kappa/2\pi$  normalization is value-inherited); the strong-field metric; the full EFE.
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## 11. Falsification Criteria

GR-III is falsifiable at every closed step and at each open number: break the lapse fork, the geodesic reduction, the khronon-speed derivation, or the field-equation/horizon build, and the corresponding result fails; the preferred-frame front closes for or against the theory once its numbers are computed. The sharpest, stated individually:

### 11.1 The lapse fork

**Falsifier sentence:** *A demonstration that the P04 commitment-reserve band co-scales with the internal band ( $\beta \neq 0$ ) — e.g. a replenishment channel slaving the reserve to  $b_{int}$  — would reopen  $\alpha \neq 1$  and the Einstein/Nordström fork, falsifying §4.*

### 11.2 The geodesic reduction

**Falsifier sentence:** *An explicit  $\Sigma$ -rule computation showing the massive-front coherence functional is **not** the Jacobi  $J(b)$  (i.e. the massive worldlines are not timelike geodesics of  $g \sim b^{-1}$ ) would falsify §5 — the limit checks (Fermat, Maupertuis) being necessary but not sufficient.*

### 11.3 The khronon speed

**Falsifier sentence:** *A demonstration that ED’s foliation/reserve sector supplies a conservative  $\lambda\theta^2$  term ( $\varepsilon \neq 0$ ) — i.e. that the reserve coupling is kinetic, not dissipative — would falsify  $c_s = c$  (§6) and give the khronon a second cone.*

### 11.4 The field equation / horizon

**Falsifier sentence:** *A demonstration that the forced rule’s steady state is **not**  $\nabla^2 b \sim \rho$ , or that strong coupling does **not** form a finite-radius  $b \rightarrow 0$  horizon / the severed entropy scales as the bulk ( $r_h^2$ ) not the surface ( $r_h$ ), would falsify §7.*

### 11.5 The preferred-frame front (when computed)

**Falsifier sentence:** *Computed  $\alpha_1$  or  $\alpha_2$  outside the lunar-laser-ranging / pulsar bounds ( $10^{-4}$ ,  $10^{-7}$ ), once  $\kappa$ ,  $D$  are pinned and the PPN expansion performed, would falsify ED-gravity-as-viable — the front this paper leaves open.*

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## 12. Position Statement

The dynamical rule **Event-Density gravity reduces to** is built and runs; its two weak-field assumptions are closed; its scalar mode is pinned to the light cone — and one irreversibility does all three structural jobs.  $\alpha = 1$  is forced from the P04 band law (the reserve cannot replenish to track the metric band — Nordström excluded); the timelike geodesic identity is reduced to the massive eikonal of GR-I’s proven null result; and the khronon propagates at  $c$  because the reserve sector is dissipative, not kinetic (KM-II §6 resolved). The rule, built, has the Newtonian field equation as its fixed point, the Schwarzschild relation  $r_s \propto M$  (2D and 3D), an emergent finite-radius horizon that is a frozen A2 cut carrying area-law entropy, and the khronometric mode count read off  $F$ . The same primitive — the arrow, P11 — selects the Einstein branch at the lapse, makes the khronon physical at the mode count, and keeps it on the light cone.

**What this paper claims.** The dynamical rule is built and Newtonian-consistent;  $\alpha = 1$  is forced (Nordström excluded); the timelike geodesic identity is limit-forced;  $c_s = c$  is derived; the emergent horizon, frozen A2 cut, and area law are measured; the mode count is from  $F$ ; and the arrow’s three structural selections are one unification.

**What this paper does not claim.** The full EFE is not derived; the class is khronometric, not pure GR;  $\alpha = 1$  rests on a band-accounting premise; the geodesic identity is reduced, not proven;  $c_s = c$  is leading-order; the rule’s coefficients are unpinned;  $\alpha_1, \alpha_2$  **are not computed (and not faked)**; the Hawking scaling is **derived** ( $\kappa = 1/(2r_s) \propto 1/r_h$ ) but its **direct** simulation is resolution-limited and its coefficient value-inherited; and Einstein is not derived.

**Series context.** Paper GR-III of the post-pivot curvature-emergence line — the engine of the quartet (GR-I, GR-II, KM-I, KM-II): the rule they reduce to, built; the assumptions they carried, closed; the scalar speed they left open, derived. Two falsification numbers remain open, stated as such.

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## Appendix: Cross-References, Glossary, Reader Map

### Cross-references

- **Paper\_087:** position paper. — **Paper\_027:** Newton’s  $G$  (value-inherited).
- **GR-I:** weak-field metric (assumptions closed here). — **GR-II:** khronometric class;  $F$ -reduction (built here).
- **KM-I/KM-II:** deep-field + cosmology (KM-II §6 resolved here).
- **Working record:** `event-density/foundations/` (the Phase-3 GR rounds and the dynamical-rule builds) and `evaluation/DynamicalBandwidth/`.

## Glossary

- **The dynamical rule**  $F$ .  $\dot{b} = F(\rho, \nabla\rho, b)$  — the bandwidth evolution whose steady state is the field equation (GR-II/R9); minimal forced form  $\dot{b} = D\nabla^2 b - \kappa\rho$ .
- **The three hats (of the arrow)**. P11 selecting the Einstein/khronometric structure three times:  $\alpha = 1$  (lapse), khronon physical (mode count), khronon at  $c$  (scalar speed).
- $\alpha = 1$  / **P-Commitment-Linear**. The lapse exponent, forced by the reserve's inability to replenish (P11); Nordström excluded.
- **Jacobi functional**.  $J(b) = b^{-1} \sqrt{(E^2 - m^2 b)}$  — the timelike-geodesic functional; limit-forced to Fermat ( $m \rightarrow 0$ ) and Maupertuis (weak field).
- $\varepsilon / c_s = c$ . The foliation-kinetic knob ( $c_s/c = \sqrt{1 + \varepsilon}$ ); derived zero (dissipative reserve), so the khronon is at light speed.
- **Frozen A2 cut**. The emergent  $b \rightarrow 0$  horizon: a metric horizon, an A2 decoupling cut, and a V5 surface — one locus.
- **Area law**  $S \propto A$ . The severed-channel count (A1) scaling as the horizon perimeter, not the bulk — holographic; coefficient value-inherited.

## Reader map and open work

**Where to look next.** - *The weak-field metric and tests*: GR-I. — *The class and GW/Lorentz*: GR-II. — *MOND/cosmology*: KM-I/KM-II. - *The build details*: `event-density/foundations/Phase3_GR_*` and `evaluation/DynamicalBandwidth/`.

**Open work** (declared): **compute**  $\alpha_1, \alpha_2$  (pin  $\kappa, D \rightarrow$  khronometric PPN, or direct PPN expansion of  $F$ ); a **fully-direct** simulation of the Hawking surface gravity (the scaling is derived; the direct measurement is resolution-limited — needs large horizons or a non-relaxational strong-field rule); the explicit  $\Sigma$ -rule massive-front computation (closes §5); the sub-leading reserve speed-renormalization; pinning  $\kappa, D$ .

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## End of Paper GR-III.

*Post-pivot curvature-emergence line, the engine of the quartet. The dynamical rule ED's gravity reduces to is built and runs (Newtonian field equation as its fixed point; Schwarzschild  $r_s \propto M$  in 2D and 3D; an emergent finite-radius horizon that is a frozen A2 cut carrying area-law entropy  $S \propto A$  and the Hawking scaling  $\kappa = 1/(2r_s) \propto 1/r_h$ ; the khronometric mode count read off  $F$ ). Its two weak-field assumptions are closed:  $\alpha = 1$  forced from the P04 band law (Nordström excluded), the timelike geodesic identity limit-forced to GR-I's proven null result. Its scalar mode is pinned to the light cone ( $c_s = c$ , derived — the reserve is dissipative, not kinetic; KM-II §6 resolved). One irreversibility (P11) does all three structural jobs — the lapse, the mode count, the scalar speed. One number stays open and is flagged:  $\alpha_1, \alpha_2$  (structurally suppressed, not computed, not faked); the Hawking scaling, earlier thought open, is corrected to a derived result (the direct sim measurement remains resolution-limited). No new primitives; Einstein not derived; the engine exists.*